Does Ukrainian serve as a good bridge language for Russian speakers in understanding other Slavic languages?

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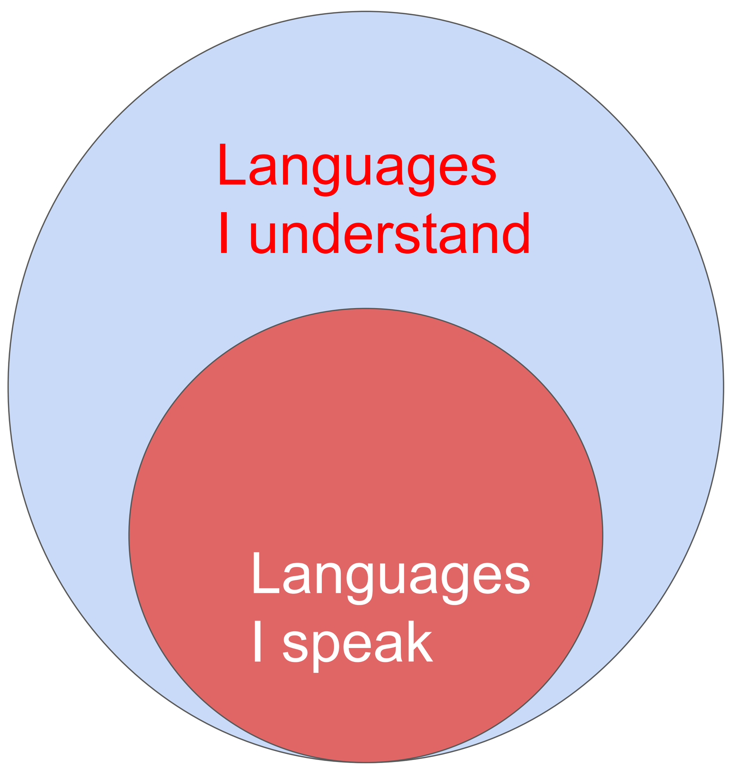
Motivation

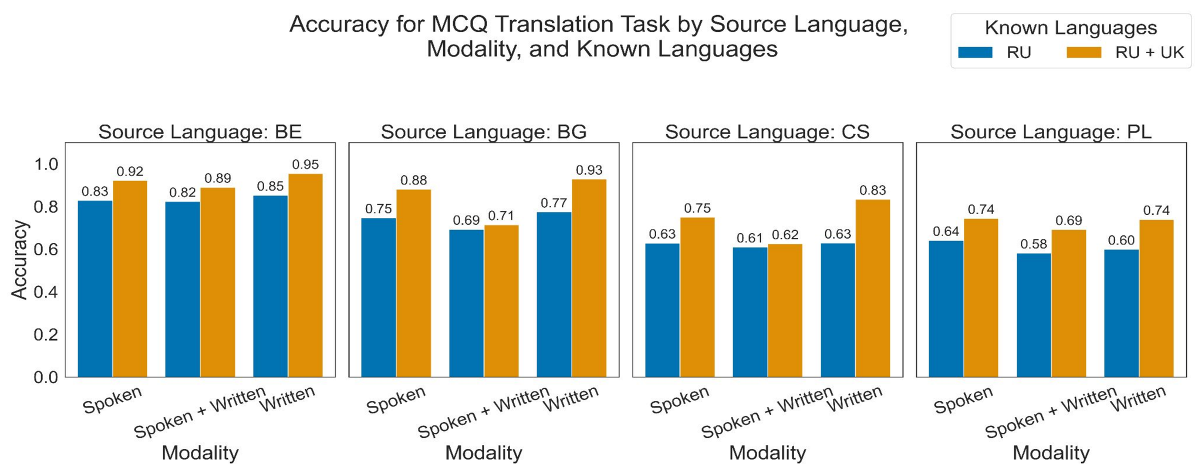


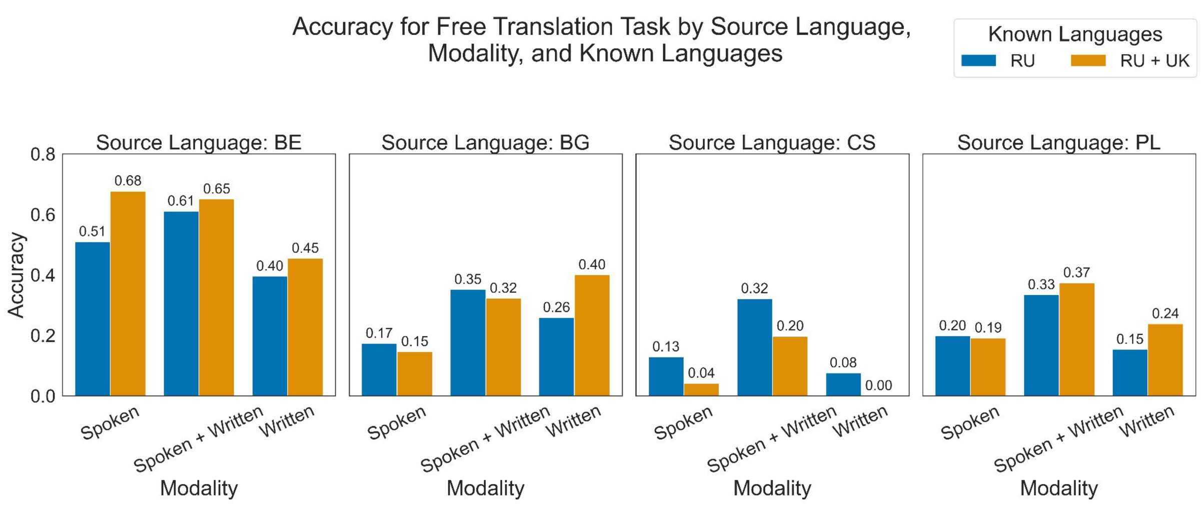
Receptive Multilingualism

- Slavic languages are typologically similar and make good candidates for examining *"Receptive Multilingualism"*.
- Can speakers use their L2 to understand Lx due to factors such as orthographic, phonological, and lexical similarities between Slavic languages?
- Ukrainian (UK) = Belarusian (BE) phonology, morphology + Czech (CS) phonology + Polish (PL) morphology, lexicon + Bulgarian (BL) and Russian (RU) lexicon → optimal Slavic bridge language

RQ1: How well do native Russian speakers with Ukrainian as L2 understand unfamiliar Slavic languages (Lx) compared to monolingual Russian speakers?

- Participants with Ukrainian knowledge tend to perform worse when translating CS during Free translation.
- Participants tend to perform best when BE is the source language.
- Higher sentence surprisal might indicate that participants are likely to engage in more careful analysis.





How Variables Affect Translation Accuracy According to Models (tested in R)

Variables	Free Translation Accuracy	MCQ Translation Accuracy
RU+UK speakers	-	↑
Source language with BE as baseline	↑	↑
Lower Levenshtein distance	↑	-
Higher Jaccard similarity	↑	↑
Interaction between RU+UK speakers and BE as a source language	↑	-
Interaction between the written modality and BG as a source language	↓	↑
Higher source phrase surprisal	↓	↓
Higher sentence level surprisal	-	↑

Experiment and Data Analysis

- Data from [1]. Two groups of participants (with and without UK knowledge) translating BE, BL, CS and PL into RU in a web-based experiment.
- Phrase to translate: *Give me at least some kind of bonus **or something**?*
- **233 RU speakers** and **20 RU+UK speakers** in the MCQ task.
- **206 RU speakers** and **20 RU+UK speakers** in the Free translation task.
- Predictor variables for the analysis:
  - Levenshtein Distance
  - Jaccard Similarity
  - LLM Surprisal of the phrase and the sentence
- Response variables for the analysis:
  - Accuracy (separate for each task)
  - COMET [2] } Translation Quality
  - BERTScore (F1-Score) [3] }

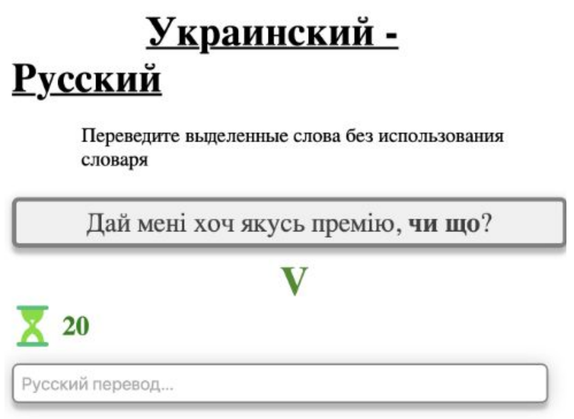


Fig 1. Free Translation Task from [1]

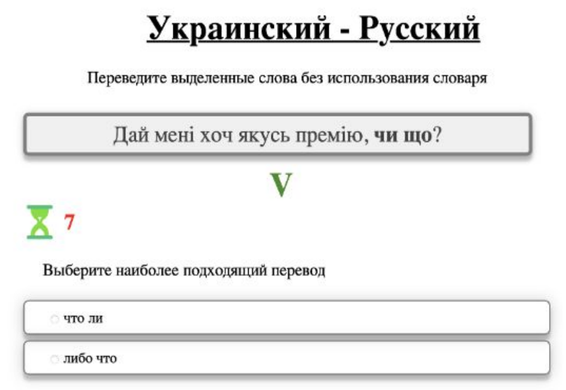
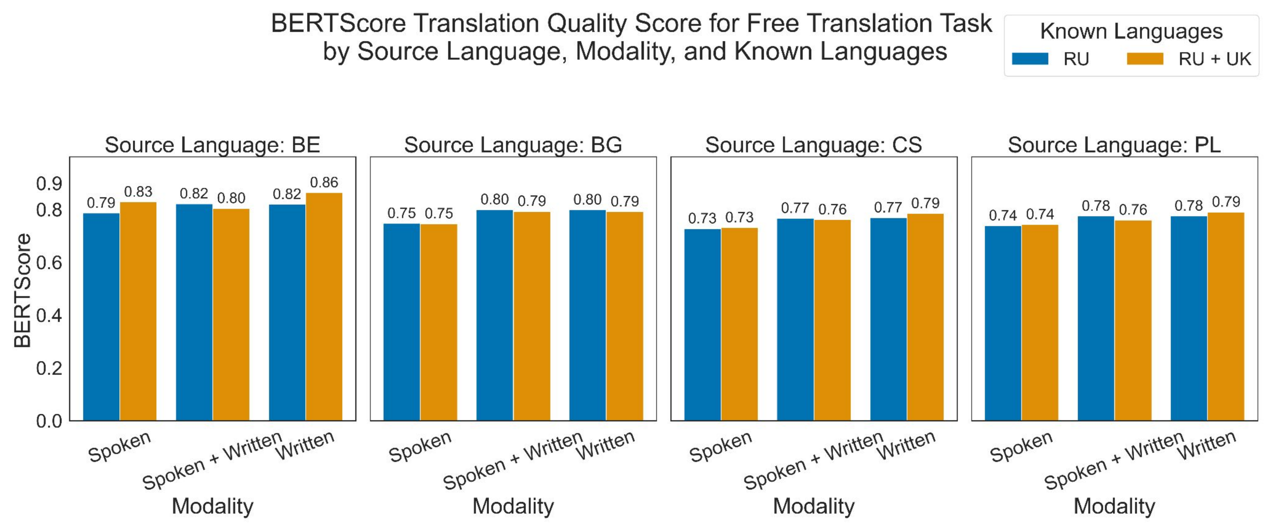
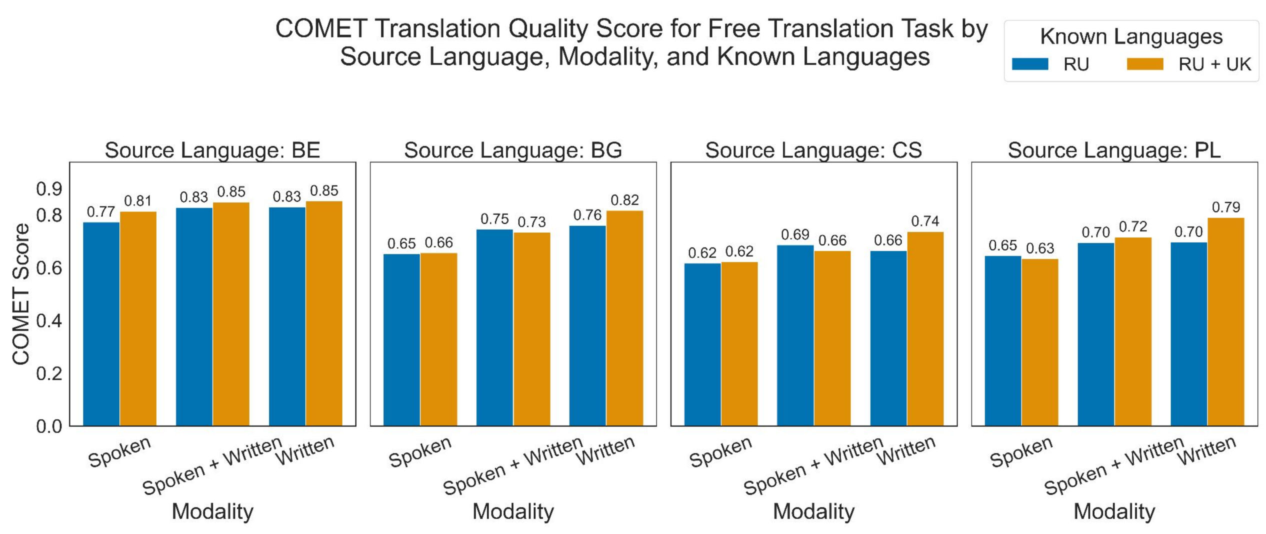


Fig 2. MCQ Translation Task from [1]

RQ2: Do these two groups differ in their translation quality, as evidenced by COMET and BERTScore?





- Participants tend to perform worse in the spoken modality.
- Performance varies by source language.
- Participants tend to perform best when BE is the source language.
- Less deviation of score in the BERTScore metric.

How Variables Affect Scores According to Models (tested in R)

Variables	COMET Score	F1-Score (from BERTScore)
RU+UK speakers	↑	↑
Lower Levenshtein distance	↑	↑
Higher Jaccard similarity	↑	↑
Source languages other than BE	↓	↓
Interaction between RU+UK speakers and BG as a source language	↓	↓
Interaction between the spoken modality and BG as a source language	↓	↓
Higher source phrase surprisal	↑	↓

Conclusion

- MCQ task: Russian speakers with Ukrainian as L2 performed better than Russian monolinguals in all four Slavic languages to varied degrees.
- Free translation task: Russian speakers with Ukrainian L2 performed worse than Russian monolinguals in Czech.
- Translation Quality: Russian speakers with Ukrainian knowledge also had better translation quality.
- Ukrainian aids Russian speakers in learning other Slavic languages but degree depends on the task, the modality, and the typological similarity.
- Limitations:
  - Imbalance in the class size leading to lower power.
  - All translations and text to be translated were transliterated into Cyrillic and this process was lossy (CS and PL diacritics were lost).

References

[1] I. Zaitova, I. Stenger, M. Butt, and T. Avgustinova. 2024. Cross-linguistic processing of non-compositional expressions in Slavic languages. In *Proceedings of the Workshop on Cognitive Aspects of the Lexicon @ LREC-COLING 2024*, pp. 86–97, Torino, Italia.

[2] R. Rei, C. Stewart, AC. Farinha, and A. Lavie. 2020. COMET: A neural framework for MT evaluation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 2685–2702, Online. Association for Computational Linguistics.

[3] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi. 2020. BERTScore: Evaluating text generation with BERT.

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